EXPRIMENT-10

Playfair Cipher Encryption

Aim:

To write a Python program to encrypt a message using the Playfair cipher with the given matrix.

PROGRAM:

main.py

```
matrix = [
    "MFHIK",
    "UNOPQ",
    "ZVWXY",
    "ELARG",
    "DSTBC"
]

def encrypt(text):
    text = text.upper().replace("J", "I")
    text = "".join(ch for ch in text if ch.isalpha())

    pairs = []
    i = 0

    while i < len(text):
        a = text[i]

        if i + 1 == len(text):
            b = 'X'
            i += 1
        elif text[i] == text[i+1]:
            b = 'X'
            i += 1
        else:
            b = text[i+1]
```

```
        for r in range(5):
            for c in range(5):
                if matrix[r][c] == a:
                    r1, c1 = r, c
                if matrix[r][c] == b:
                    r2, c2 = r, c

        if r1 == r2:
            result += matrix[r1][(c1+1)%5]
            result += matrix[r2][(c2+1)%5]

        elif c1 == c2:
            result += matrix[(r1+1)%5][c1]
            result += matrix[(r2+1)%5][c2]

        else:
            result += matrix[r1][c2]
            result += matrix[r2][c1]

    return result


message = "Must see you over Cadogan West. Coming at once."

print("Plain Text:", message)
print("Cipher Text:", encrypt(message))
```

OUTPUT:

```
Plain Text: Must see you over Cadogan West. Coming at once.
Cipher Text: UZTBDLGZPNNWLGTGTUEROVLDBDUHFPERHWQSRZ
```

RESULT:

Thus, the given plaintext was successfully encrypted using the Playfair cipher.